

AWS Cloud Security Assessment Report Using Prowler

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Assessment Tool: Prowler

Cloud Platform: Amazon Web Services

1. Executive Summary

This report presents the results of a cloud security assessment conducted on an AWS environment using the automated auditing tool Prowler. The objective of the assessment was to evaluate the security posture of the environment and identify misconfigurations that could expose cloud resources to potential threats.

The security scan evaluated multiple AWS services including identity management, monitoring services, network configurations, and encryption settings. The assessment was performed using an IAM audit account configured through the AWS CLI on a Kali Linux environment.

The scan executed **142 security checks**, out of which **64 passed while 78 failed**, indicating that several security configurations require remediation to align with industry best practices. Key areas with the highest number of failed checks include:

- Identity and access management configurations
- Cloud monitoring and logging settings
- Network access controls
- Encryption configurations

These findings highlight the need for stronger identity protection, improved monitoring mechanisms, and stricter network access controls within the AWS environment.

2. Objectives

The objectives of this project were to:

- Assess the security posture of a cloud infrastructure hosted on AWS.
- Identify potential vulnerabilities and misconfigurations across AWS services.
- Evaluate compliance with cloud security best practices.
- Provide remediation recommendations to improve the security posture of the environment.

3. Scope of Assessment

The security assessment covered several AWS services within the account.

Services assessed include:

- AWS Identity and Access Management
- Amazon EC2
- Amazon CloudWatch
- AWS Config
- Amazon VPC
- AWS Glue
- AWS Organizations
- Amazon S3

The scan was performed in the **us-east-1 region** of the AWS environment.

4. Methodology

The following methodology was used to conduct the assessment.

4.1 Environment Preparation

An IAM user with the required permissions was created within the AWS account to perform security auditing tasks.

The user credentials were configured using the AWS CLI on a Kali Linux environment.

4.2 AWS CLI Configuration

The AWS CLI was configured with the audit profile:

```
aws configure --profile blu_prowler_user
```

Authentication was verified using:

```
aws sts get-caller-identity --profile blu_prowler_user
```

4.3 Security Scan Execution

The security assessment was conducted using the command:

```
prowler aws --profile blu_prowler_user --region us-east-1
```

4.4 Results Analysis

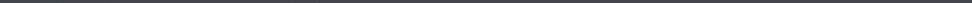
The scan results were analyzed to identify failed checks and categorize them by affected AWS services. The Prowler terminal summary provided a high-level overview of failed checks across services.

5. Scan Summary

The Prowler scan produced the following overall results:

Session Actions Edit View Help						
52.35% (78) Failed 42.95% (64) Passed						
Account 386397332453 Scan Results (severity columns are for fails only):						
Provider	Service	Status	Critical	High	Medium	Low
aws	accessanalyzer	FAIL (1)	0	0	0	1
aws	account	FAIL (1)	0	0	1	0
aws	athena	FAIL (2)	0	0	2	0
aws	backup	FAIL (1)	0	0	0	1
aws	cloudtrail	FAIL (4)	0	0	0	4
aws	cloudwatch	FAIL (15)	0	0	15	0
aws	config	FAIL (1)	0	0	1	0
aws	ec2	FAIL (5)	0	1	4	0
aws	fms	PASS (1)	0	0	0	0
aws	glue	FAIL (2)	0	0	2	0
aws	iam	FAIL (36)	0	4	15	11
aws	network-firewall	FAIL (1)	0	0	1	0
aws	organizations	FAIL (2)	0	0	2	1
aws	resourceexplorer2	PASS (1)	0	0	0	0
aws	s3	FAIL (1)	0	1	0	0
aws	ssm	FAIL (1)	0	0	0	1
aws	trustedadvisor	PASS (1)	0	0	0	0
aws	support	FAIL (1)	0	0	0	1
aws	vpc	FAIL (9)	0	0	9	0
* You only see here those services that contains resources.						
Detailed results are in:						
- HTML: /home/kali/cloudvuln/prowler/permissions/output/prowler-output-386397332453-20260312100523.html						
- JSON-OCF: /home/kali/cloudvuln/prowler/permissions/output/prowler-output-386397332453-20260312100523.ocsf.json						
- CSV: /home/kali/cloudvuln/prowler/permissions/output/prowler-output-386397332453-20260312100523.csv						
- JSON: /home/kali/cloudvuln/prowler/permissions/output/prowler-output-386397332453-20260312100523.json						

Report Information											
Version: 3.11.3											
Parameters used: aws -profile blu_prowler_user -region us-east-1											
Date: 2026-03-12T10:05:23.742594											
AWS Assessment Summary											
AWS Account: 386397332453											
AWS-CLI Profile: blu_prowler_user											
Audited Regions: us-east-1											
AWS Credentials											
User Id: AIDAVTSYTMP5RPGNPXCR											
Caller Identity ARN: arn:aws:iam:386397332453:user:prowlerscan_user											
Assessment Overview											
Total Findings: 142											
Passed: 64											
Failed: 78											
Total Resources: 41											
Search:											
Filters Show 100 entries											
Status	Severity	Service Name	Region	Check ID	Check Title	Resource ID	Resource Tags	Status Extended	Risk	Recommendation	Compliance
FAIL	low	support	us-east-1	trustedadvisor_premium_support_plan_subscribed	Check if a Premium support plan is subscribed	386397332453		Amazon Web Services Premium Support Plan isn't subscribed.	Ensure that the appropriate su read more...	It is recommended that you sub read more...	
FAIL	medium	organizations	us-east-1	organizations_account_part_of_organizations	Check if account is part of an AWS Organizations	AWS Organization		AWS Organizations is not in-use for this AWS Account.	The risk associated with not b read more...	Create or Join an AWS Organiza read more...	-MITRE-ATTACK: T1078, T1087, read more...
FAIL	medium	organizations	us-east-1	organizations_tags_policies_enabled_and_attached	Check if an AWS Organization has tags policies enabled and attached.	AWS Organization		AWS Organizations is not in-use for this AWS Account.	If an AWS Organization tags po read more...	Enable and attach AWS Organiza read more...	-ENS-RD2022: op.exp.1.aws.sys read more...
FAIL	low	organizations	us-east-1	organizations_scp_check_deny_regions	Check if AWS Regions are restricted with SCP policies	AWS Organization		AWS Organizations is not in-use for this AWS Account.	The risk associated with not r read more...	Restrict AWS Regions using SCP read more...	-MITRE-ATTACK: T1078, T1535 read more...
PASS	medium	cloudwatch	us-east-1	cloudwatch_cross_account	Check if CloudWatch has allowed	386397332453		CloudWatch doesn't allow cross-account	Cross-	Grant usage permission on a	-AWS-Well-



The screenshot shows a web browser window with the address bar displaying the file path: `file:///home/kali/cloudvuln/prowler/permissions/output/prowler-output-386397332453-20260312100523.html`. The browser tabs include 'GitHub - prowler-cloud/prowler' and 'Prowler - The Handy Cloud S...'. The taskbar at the bottom shows various Kali Linux tools and forums.

	Report Information Version: 3.11.3 Parameters used: aws --profile blu_prowler_user --region us-east-1 Date: 2020-09-12T13:05:23.742004	AWS Assessment Summary AWS Account: 390397323453 AWS CLI Profile: blu_prowler_user Assessed Regions: us-east-1	AWS Credentials User ARN: arn:aws:iam::390397323453:user/blu_prowler_user Caller Identity ARN: arn:aws:iam::390397323453:user/blu_prowler_user	Assessment Overview Total Findings: 142 Passed: 54 Failed: 75 Total Resources: 41
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Status	Severity	Service Name	Region	Check ID	Check Title	Resource ID	Resource Type	Status Extended	Risk	Recommendation	Compliance
PA05	critical	iam	us-east-1	iam_not_mfa_enabled	Ensure MFA is enabled for the root account	root_account	Root account	MFA is enabled for root account	The root account is the most privileged user in an AWS account. MFA adds an extra layer of protection on top of a user name and password, with MFA enabled when a user logs in to the AWS console. MFA adds an extra layer of protection on top of a user name and password, with MFA enabled when a user logs in to the AWS console. MFA adds an extra layer of protection on top of a user name and password, with MFA enabled when a user logs in to the AWS console.	Using IAM console navigate to root user .	<COP2 article_25-GSP-21 CF root more
PA05	critical	iam	us-east-1	iam_not_root_account_key	Ensure no root account access key exists	root_account	Root account	root_account does not have access keys.	The root account is the most privileged user in an AWS account. MFA adds an extra layer of protection on top of a user name and password, with MFA enabled when a user logs in to the AWS console. MFA adds an extra layer of protection on top of a user name and password, with MFA enabled when a user logs in to the AWS console. MFA adds an extra layer of protection on top of a user name and password, with MFA enabled when a user logs in to the AWS console.	Use the console to go to root more .	<COP2 article_25-GSP-21 CF root more
PA05	critical	iam	us-east-1	iam_not_hardware_mfa_enabled	Ensure only hardware MFA is enabled for the root account	root_account	Root account	Root account has a hardware MFA device enabled	The root account is the most privileged user in an AWS account. MFA adds an extra layer of protection on top of a user name and password, with MFA enabled when a user logs in to the AWS console. MFA adds an extra layer of protection on top of a user name and password, with MFA enabled when a user logs in to the AWS console. MFA adds an extra layer of protection on top of a user name and password, with MFA enabled when a user logs in to the AWS console.	Using IAM console navigate to root user and root user .	<COP2 article_25-GSP-21 CF root more

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Product	Severity	Category	Control ID	Control Description	Control Rationale	Control Implementation	Control Evidence	Control Status	Control Comments	
PAWS	high	sen	LO-4440-1	sen_customer_attached_policy_m Administrative privileges	Ensure IAM Customer-Managed policies that allow full temp "*/" administrative privileges are not attached	Customer policy temp is disabled but does not allow "*/" administrative privileges	sen policies are the means by read more...	It is more secure to start with read more...	HCSPY: AWS::IAM::Group-21 CI read more...	
PAWS	high	sen	LO-4440-2	sen_cis_ssm_session_restricted_deploy	Ensure role Service Role prevents against a cross-service console deploy attack	IAM Service Role and S3Access does not prevent against a cross-service console deploy attack	Allow attackers to gain cross-aws read more...	Use the aws:touchpoint and aws: read more...	HCSPY-ATTACK: T1078, AWS::IAM::Role read more...	
PAWS	high	sen	LO-4440-3	sen_policy_policy_m Administrative privileges	Ensure role policies that allow full "*/" administrative privileges are not associated to vital identities	iam:policy:policy for iam identity are not associated to vital identities	iam:policy:policy for iam identity are not associated to vital identities	It is more secure to start with read more...	HCSPY: AWS::IAM::Group-21 CI read more...	
PAWS	high	sen	LO-4440-4	sen_policy_policy_m Administrative privileges	Ensure role policies that allow full "*/" administrative privileges are not associated to vital identities	iam:policy:policy for iam identity are not associated to vital identities	iam:policy:policy for iam identity are not associated to vital identities	It is more secure to start with read more...	HCSPY: AWS::IAM::Group-21 CI read more...	
PAWS	high	sen	LO-4440-5	sen_user_mfa_enabled_console_access	Ensure multi-factor authentication (MFA) is enabled for all vital users that have a console password	User root_account has Console Password enabled and MFA enabled	Unauthenticated access to AWS read more...	Enable MFA for the user's access read more...	HCSPY: AWS::IAM::Group-21 CI read more...	
PAWS	high	sen	LO-4440-6	sen_user_mfa_enabled_console_access	Ensure multi-factor authentication (MFA) is enabled for all vital users that have a console password	User jason has Console Password enabled and MFA enabled	Unauthenticated access to AWS read more...	Enable MFA for the user's access read more...	HCSPY: AWS::IAM::Group-21 CI read more...	
PAWS	high	sen	LO-4440-7	sen_user_mfa_enabled_console_access	Ensure multi-factor authentication (MFA) is enabled for all vital users that have a console password	User jason has Console Password enabled but MFA disabled	Unauthenticated access to AWS read more...	Enable MFA for the user's access read more...	HCSPY: AWS::IAM::Group-21 CI read more...	
PAWS	high	sen	LO-4440-8	sen_user_mfa_enabled_console_access	Ensure multi-factor authentication (MFA) is enabled for all vital users that have a console password	User jason has Console Password enabled but MFA disabled	Unauthenticated access to AWS read more...	Enable MFA for the user's access read more...	HCSPY: AWS::IAM::Group-21 CI read more...	
PAWS	high	sen	LO-4440-9	sen_policy_admin_privilege_execution	Ensure no Customer Managed IAM policies allow actions that may lead to privilege escalation	temp	Custom Policy iam:arn:aws:iam::38637310343:policytemp does not allow privilege escalation	Users with some IAM permission read more...	Grant usage permission on a per read more...	HCSPY-ATTACK: T1078, T1048, read more...
PAWS	high	sen	LO-4440-10	sen_policy_admin_privilege_execution	Ensure no Customer Managed IAM policies allow actions that may lead to privilege escalation	Access	Custom Policy iam:arn:aws:iam::38637310343:policytemp does not allow privilege escalation	Users with some IAM permission read more...	Grant usage permission on a per read more...	HCSPY-ATTACK: T1078, T1048, read more...
PAWS	high	sen	LO-4440-11	sen_securitygroup_allow_ingress_from_internet_to_any_port	Ensure no security groups allow ingress from 0.0.0.0/0 - 0.0.0.0/0 to any port	sg-000000001760223a	Security group default (sg-000000001760223a) does not have all ports open to the internet	If security groups are not per read more...	<CIS 1.4.2 <CIS> you may read more...	
PAWS	high	sen	LO-4440-12	sen_securitygroup_allow_ingress_from_internet_to_tcp_cassandra_7135_7200_8000	Ensure no security groups allow ingress from 0.0.0.0/0 - 0.0.0.0/0 to Cassandra ports 7135 or 7200 or 8000	sg-000000001760223a	Security group default (sg-000000001760223a) does not have Cassandra ports 7135, 8000 and 7200 open to the internet	If security groups are not properly configured and the attack surface is increased	HCSPY-ATTACK: T1198, T1048, T1049, T1046, T1047 read more...	
PAWS	high	sen	LO-4440-13	sen_securitygroup_allow_ingress_from_internet_to_tcp_elasticsearch_9200_9300	Ensure no security groups allow ingress from 0.0.0.0/0 - 0.0.0.0/0 to Elasticsearch ports 9200 or 9300	sg-000000001760223a	Security group default (sg-000000001760223a) does not have Elasticsearch ports 9200, 9300 and 9301 open to the internet	If security groups are not properly configured and the attack surface is increased	HCSPY-ATTACK: T1198, T1048, T1049, T1046, T1047 read more...	
PAWS	high	sen	LO-4440-14	sen_securitygroup_allow_ingress_from_internet_to_tcp_elasticsearch_9200_9300	Ensure no security groups allow ingress from 0.0.0.0/0 - 0.0.0.0/0 to Elasticsearch ports 9200 or 9300	sg-000000001760223a	Security group default (sg-000000001760223a) does not have Elasticsearch ports 9200, 9300 and 9301 open to the internet	If security groups are not properly configured and the attack surface is increased	HCSPY-ATTACK: T1198, T1048, T1049, T1046, T1047 read more...	

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Showing 1 to 40 of 40 entries (filtered from 249 total entries)									
FAIL	low	support	us-east-1	trustedadvisor_premium_support_plan_subscribed	Check if a Premium support plan is subscribed	38639732463	Amazon Web Services Premium Support Plan isn't subscribed	Risk	Ensure that the appropriate support plan is subscribed
FAIL	low	organizations	us-east-1	organizations_scp_check_demo_regions	Check if AWS Regions are restricted with SCP policies	AWS Organization	AWS Organizations is not in use for this AWS Account	The risk associated with not using SCP policies is increased	Prohibit AWS Regions using SCP policies
FAIL	low	accessanalyzer	us-east-1	accessanalyzer_enabled	Check if IAM Access Analyzer is enabled	38639732463	IAM Access Analyzer in account 38639732463 is not enabled	AWS IAM Access Analyzer helps read more...	Enable IAM Access Analyzer for read more...
FAIL	low	cloudtrail	us-east-1	cloudtrail_s3_dataevents_read_enabled	Check if S3 buckets have Object level logging for read events is enabled in CloudTrail	38639732463	No CloudTrail trails have a data event to record all S3 object-level API operations	If logs are not enabled, read more...	Enable logs. Create an S3 life read more...
FAIL	low	cloudtrail	us-east-1	cloudtrail_s3_dataevents_write_enabled	Check if S3 buckets have Object level logging for write events is enabled in CloudTrail	38639732463	No CloudTrail trails have a data event to record all S3 object-level API operations	If logs are not enabled, read more...	Enable logs. Create an S3 life read more...
FAIL	low	iam	us-east-1	iam_securityaudit_sfn_created	Ensure a Security Audit rule has been created to conduct security audits	SecurityAudit	SecurityAudit policy is not attached to any role	Creating an IAM role with a role read more...	Create an IAM role for conduct read more...
FAIL	low	backup	us-east-1	backup_vaults_exist	Ensure AWS Backup vaults exist	38639732463	No Backup Vault exist	Without an AWS Backup vault, a read more...	Use AWS Backup to create backup read more...
FAIL	low	cloudtrail	us-east-1	cloudtrail_insights_enabled	Ensure CloudTrail Insights is enabled	documentation:trails	Trail documentation trails do not have insight selectors and it is logging	CloudTrail Insights provides a read more...	Enable CloudTrail Insights read more...
PASS	low	cloudtrail	us-east-1	cloudtrail_multi_region_enabled_logging_management_events	Ensure CloudTrail logging management events in all Regions	documentation:trails	Trail documentation trails from home region us-east-1 to multi-region, in logging and trace management events enabled	AWS CloudTrail enables governa read more...	Enable CloudTrail logging manag read more...
FAIL	low	cloudtrail	us-east-1	cloudtrail_cloudwatch_logging_enabled	Ensure CloudTrail trails are integrated with CloudWatch Logs	documentation:trails	Multi-region trail documentation trails is not logging in the last 24h or not configured to deliver logs	Validate that the trails in CI read more...	<COPR, article_26, article_30 read more...
FAIL	low	iam	us-east-1	iam_policy_attached_only_to_group_or_roles	Ensure IAM policies are attached only to groups or roles	MultiUserAccess	User blu has the policy AdministratorAccess attached	By default IAM users, groups, read more...	Remove any policy attached de read more...
FAIL	low	iam	us-east-1	iam_policy_attached_only_to_group_or_roles	Ensure IAM policies are attached only to groups or roles	MultiUserChangePassword	User blu has the policy IAMUserChangePassword attached	By default IAM users, groups, read more...	Remove any policy attached de read more...
FAIL	low	iam	us-east-1	iam_policy_attached_only_to_group_or_roles	Ensure IAM policies are attached only to groups or roles	jumpstack	User jason has the policy temp attached	By default IAM users, groups, read more...	Remove any policy attached de read more...

FAIL	low	cloudtrail	us-east-1	cloudtrail_multi_region_enabled_logging_management_events	Ensure CloudTrail logging management events in all Regions	documentation:trails	Trail documentation trails from home region us-east-1 to multi-region, in logging and trace management events enabled	AWS CloudTrail enables governa read more...	Enable CloudTrail logging manag read more...
FAIL	low	cloudtrail	us-east-1	cloudtrail_cloudwatch_logging_enabled	Ensure CloudTrail trails are integrated with CloudWatch Logs	documentation:trails	Multi-region trail documentation trails is not logging in the last 24h or not configured to deliver logs	Validate that the trails in CI read more...	<COPR, article_26, article_30 read more...
FAIL	low	iam	us-east-1	iam_policy_attached_only_to_group_or_roles	Ensure IAM policies are attached only to groups or roles	MultiUserAccess	User blu has the policy AdministratorAccess attached	By default IAM users, groups, read more...	Remove any policy attached de read more...
FAIL	low	iam	us-east-1	iam_policy_attached_only_to_group_or_roles	Ensure IAM policies are attached only to groups or roles	MultiUserChangePassword	User blu has the policy IAMUserChangePassword attached	By default IAM users, groups, read more...	Remove any policy attached de read more...
FAIL	low	iam	us-east-1	iam_policy_attached_only_to_group_or_roles	Ensure IAM policies are attached only to groups or roles	jumpstack	User jason has the policy temp attached	By default IAM users, groups, read more...	Remove any policy attached de read more...

FAIL	low	iam	us-east-1	iam_policy_attached_only_to_group_or_roles	Ensure IAM policies are attached only to groups or roles	MuhammadUserChangePassword	User blu has the policy IAMUserChangePassword attached.	By default IAM users, groups, read more...	Remove any policy attached to read more...	<GP-25-CFR-Part 11: 11.10-6, read more...
FAIL	low	iam	us-east-1	iam_policy_attached_only_to_group_or_roles	Ensure IAM policies are attached only to groups or roles	jeanloup	User jeanloup has the policy temp attached.	By default IAM users, groups, read more...	Remove any policy attached to read more...	<GP-25-CFR-Part 11: 11.10-6, read more...
FAIL	low	iam	us-east-1	iam_policy_attached_only_to_group_or_roles	Ensure IAM policies are attached only to groups or roles	jeanloupViewOnlyAccess	User jeanloup has the policy ViewOnlyAccess attached.	By default IAM users, groups, read more...	Remove any policy attached to read more...	<GP-25-CFR-Part 11: 11.10-6, read more...
FAIL	low	iam	us-east-1	iam_policy_attached_only_to_group_or_roles	Ensure IAM policies are attached only to groups or roles	jeanloupSecurityAudit	User jeanloup has the policy SecurityAudit attached.	By default IAM users, groups, read more...	Remove any policy attached to read more...	<GP-25-CFR-Part 11: 11.10-6, read more...
FAIL	low	iam	us-east-1	iam_policy_attached_only_to_group_or_roles	Ensure IAM policies are attached only to groups or roles	jeanloupProwlerPolicy	User jeanloup has the inline policy prowler-policy attached.	By default IAM users, groups, read more...	Remove any policy attached to read more...	<GP-25-CFR-Part 11: 11.10-6, read more...
FAIL	low	iam	us-east-1	iam_policy_attached_only_to_group_or_roles	Ensure IAM policies are attached only to groups or roles	prowlersec_userViewOnlyAccess	User prowlersec_user has the policy ViewOnlyAccess attached.	By default IAM users, groups, read more...	Remove any policy attached to read more...	<GP-25-CFR-Part 11: 11.10-6, read more...
FAIL	low	iam	us-east-1	iam_policy_attached_only_to_group_or_roles	Ensure IAM policies are attached only to groups or roles	prowlersec_userSecurityAudit	User prowlersec_user has the policy SecurityAudit attached.	By default IAM users, groups, read more...	Remove any policy attached to read more...	<GP-25-CFR-Part 11: 11.10-6, read more...
FAIL	low	iam	us-east-1	iam_policy_attached_only_to_group_or_roles	Ensure IAM policies are attached only to groups or roles	prowlersec_userIAMUserChangePassword	User prowlersec_user has the policy IAMUserChangePassword attached.	By default IAM users, groups, read more...	Remove any policy attached to read more...	<GP-25-CFR-Part 11: 11.10-6, read more...
FAIL	low	iam	us-east-1	iam_policy_attached_only_to_group_or_roles	Ensure IAM policies are attached only to groups or roles	prowlersec_userProwlerPolicy	User prowlersec_user has the inline policy prowler_policy attached.	By default IAM users, groups, read more...	Remove any policy attached to read more...	<GP-25-CFR-Part 11: 11.10-6, read more...
PASS	low	iam	us-east-1	iam_customer_unattached_policy_no_administrative_privileges	Ensure IAM policies that allow full *** administrative privileges are not created	Access	Custom policy Access is unattached and does not allow *** administrative privileges.	IAM policies are the means by read more...	It is more secure to start with read more...	<HTTR-ATTACK: T1078-WNS-WNS, read more...
FAIL	low	iam	us-east-1	saml_credentials_enabled_with_plans	Ensure SSM Incidents is enabled with response plans	38630732453	No SSM Incidents replication set exists.	Not having SSM Incidents enabled can plans increase the risk of delayed detection and response to security incidents, unauthorized access, limited visibility into incidents and vulnerabilities	Enable SSM Incidents and create response plans	<WS-RD3022: op-exp.8 aws.ins.1
PASS	low	resourceexplorer2	eu-north-1	resourceexplorer2_indexes_bound	Resource Explorer indexes bound	38630732453	Resource Explorer indexes bound: 2	Not having Resource Explorer indexes can result in increased complexity and overhead in managing your resources, as well as increased risk of security and compliance issues.	Create indexes	<WS-RD3022: op-exp.1 aws.ins.1

Showing 1 to 23 of 23 entries (filtered from 148 total entries)

FAIL	medium	organizations	us-east-1	organizations_account_part_of_organizations	Check if account is part of an AWS Organizations	AWS Organization	AWS Organizations is not in use for this AWS Account.	The risk associated with not being a member of an AWS Organization is increased complexity and overhead in managing your resources, as well as increased risk of security and compliance issues.	Create or join an AWS Organization	<HTTR-ATTACK: T1078, T1087, read more...
FAIL	medium	organizations	us-east-1	organizations_tag_policy_enabled_and_attached	Check if an AWS Organization has tag policies enabled and attached.	AWS Organization	AWS Organizations is not in use for this AWS Account.	If an AWS Organization tag policy is not attached, it increases the risk of unauthorized access, limited visibility into incidents and vulnerabilities.	Enable and attach AWS Organization tag policy	<WS-RD3022: op-exp.1 aws.org, read more...
PASS	medium	cloudwatch	us-east-1	cloudwatch_cross_account_sharing_disabled	Check if CloudWatch has allowed cross-account sharing	38630732453	CloudWatch doesn't allow cross-account sharing	Cross Account access to CloudWatch increases the risk of unauthorized access, limited visibility into incidents and vulnerabilities.	Grant usage permission on a per-account basis	<WS-RD3022: op-exp.1 aws.ins.1
FAIL	medium	ec2	us-east-1	ec2_its_default_encryption	Check if EC2 Default Encryption is activated.	38630732453	EC2 Default Encryption is not activated.	If not enabled sensitive info can be leaked.	Enable Encryption. Use a CMK in read more...	<GP-25-CFR-Part 11: 11.10-6, read more...
FAIL	medium	glue	us-east-1	glue_data_catalog_connection_passwords_encryption_enabled	Check if Glue data catalog settings have encrypt connection password enabled.	38630732453	Glue data catalog connection password is not encrypted.	If not enabled sensitive info can be leaked.	On the AWS Glue console, you can read more...	<WS-RD3022: op-exp.1 aws.ins.1
FAIL	medium	glue	us-east-1	glue_data_catalog_instance_encryption_enabled	Check if Glue data catalog settings have metadata encryption enabled.	38630732453	Glue data catalog settings have metadata encryption disabled.	If not enabled sensitive info can be leaked.	Enable Encryption. Use a CMK in read more...	<WS-RD3022: op-exp.1 aws.ins.1
FAIL	medium	iam	us-east-1	iam_user_hardware_mfa_enabled	Check if IAM users have Hardware MFA enabled.	blu	User blu has a virtual MFA instead of a hardware MFA device enabled.	Hardware MFA is preferred over software MFA.	Enable Hardware MFA device for read more...	<WS-RD3022: op-exp.1 aws.ins.1
FAIL	medium	iam	us-east-1	iam_user_hardware_mfa_enabled	Check if IAM users have Hardware MFA enabled.	jeanloup	User jeanloup does not have any type of MFA enabled.	Hardware MFA is preferred over software MFA.	Enable Hardware MFA device for read more...	<WS-RD3022: op-exp.1 aws.ins.1
FAIL	medium	iam	us-east-1	iam_user_hardware_mfa_enabled	Check if IAM users have Hardware MFA enabled.	prowlersec_user	User prowlersec_user does not have any type of MFA enabled.	Hardware MFA is preferred over software MFA.	Enable Hardware MFA device for read more...	<WS-RD3022: op-exp.1 aws.ins.1
PASS	medium	iam	us-east-1	iam_user_max_active_sessions_key	Check if IAM users have two active sessions keys	<root_account>	User <root_account> does not have 2 active access keys.	Access keys could be lost or a read more...	Avoid using long lived access keys	<WS-RD3022: op-exp.1 aws.ins.1
PASS	medium	iam	us-east-1	iam_user_max_active_sessions_key	Check if IAM users have two active sessions keys	blu	User blu does not have 2 active access keys.	Access keys could be lost or a read more...	Avoid using long lived access keys	<WS-RD3022: op-exp.1 aws.ins.1
PASS	medium	iam	us-east-1	iam_user_max_active_sessions_key	Check if IAM users have two active sessions keys	jeanloup	User jeanloup does not have 2 active access keys.	Access keys could be lost or a read more...	Avoid using long lived access keys	<WS-RD3022: op-exp.1 aws.ins.1
FAIL	medium	iam	us-east-1	iam_user_max_active_sessions_key	Check if IAM users have two active sessions keys	prowlersec_user	User prowlersec_user has 2 active access keys.	Access keys could be lost or a read more...	Avoid using long lived access keys	<WS-RD3022: op-exp.1 aws.ins.1

FAIL	medium	iam	us-east-1	iam_user_mf_setup_access_key	Check if IAM users have active access keys	profreader_user	User profreader_user has 2 active access keys.	Access keys could be rotated using long lived access keys or a read more...	<CIS 1.4.1.3.3 MITRE-ATTACK read more...
INFO	medium	trustedadvice	us-east-1	trustedadvice_aws_and_awsinsights	Check Trusted Advisor for errors and warnings.	38639732453	Amazon Web Services Trusted Advisor/Insights is required to use this service.	improve the security of your read more...	review and act upon to recover read more...
PASS	medium	iam	us-east-1	iam_user_mf_setup_initial_access_key	Do not setup access keys during initial user setup for all IAM users that have a console password	<root_account>	User root_account does not have access keys or uses the access keys configured.	AWS console defaults From the IAM console, generate read more...	<CIS 1.4.1.3.3 MITRE-ATTACK read more...
PASS	medium	iam	us-east-1	iam_user_mf_setup_initial_access_key	Do not setup access keys during initial user setup for all IAM users that have a console password	file	User file does not have access keys or uses the access keys configured.	AWS console defaults From the IAM console, generate read more...	<CIS 1.4.1.3.3 MITRE-ATTACK read more...
PASS	medium	iam	us-east-1	iam_user_mf_setup_initial_access_key	Do not setup access keys during initial user setup for all IAM users that have a console password	jean	User jean does not have access keys or uses the access keys configured.	AWS console defaults From the IAM console, generate read more...	<CIS 1.4.1.3.3 MITRE-ATTACK read more...
FAIL	medium	iam	us-east-1	iam_user_mf_setup_initial_access_key	Do not setup access keys during initial user setup for all IAM users that have a console password	profreader_user	User profreader_user has never used access key 1.	AWS console defaults From the IAM console, generate read more...	<CIS 1.4.1.3.3 MITRE-ATTACK read more...
FAIL	medium	iam	us-east-1	iam_user_mf_setup_initial_access_key	Do not setup access keys during initial user setup for all IAM users that have a console password	profreader_user	User profreader_user has never used access key 2.	AWS console defaults From the IAM console, generate read more...	<CIS 1.4.1.3.3 MITRE-ATTACK read more...
FAIL	medium	cloudwatch	us-east-1	cloudwatch_log_metrics_filter_and_alarm_for_aws_config_configuration_changes_enabled	Ensure a log metric filter and alarm exist for aws_config_configuration_changes_enabled	38639732453	No CloudWatch log groups found with metric filters or alarms associated.	Monitoring unauthenticated API call read more...	It is recommended that a metric read more... <CIS 1.4.1.3.3 MITRE-ATTACK read more... <CIS 1.4.1.3.3 MITRE-ATTACK read more...
FAIL	medium	cloudwatch	us-east-1	cloudwatch_log_metrics_filter_authentication_failures	Ensure a log metric filter and alarm exist for aws_managed_console_authentication_failures	38639732453	No CloudWatch log groups found with metric filters or alarms associated.	Monitoring unauthenticated API call read more...	It is recommended that a metric read more... <CIS 1.4.1.3.3 MITRE-ATTACK read more... <CIS 1.4.1.3.3 MITRE-ATTACK read more...
FAIL	medium	cloudwatch	us-east-1	cloudwatch_log_metrics_filter_iam_organizations_changes	Ensure a log metric filter and alarm exist for aws_organizations_changes	38639732453	No CloudWatch log groups found with metric filters or alarms associated.	Monitoring unauthenticated API call read more...	It is recommended that a metric read more... <CIS 1.4.1.3.3 MITRE-ATTACK read more... <CIS 1.4.1.3.3 MITRE-ATTACK read more...
FAIL	medium	cloudwatch	us-east-1	cloudwatch_changes_to_network_acl_alarm_configured	Ensure a log metric filter and alarm exist for changes to network access control lists (NACLs)	38639732453	No CloudWatch log groups found with metric filters or alarms associated.	Monitoring unauthenticated API call read more...	It is recommended that a metric read more... <CIS 1.4.1.3.3 MITRE-ATTACK read more... <CIS 1.4.1.3.3 MITRE-ATTACK read more...
FAIL	medium	cloudwatch	us-east-1	cloudwatch_changes_to_network_gateways_alarm_configured	Ensure a log metric filter and alarm exist for changes to network gateways	38639732453	No CloudWatch log groups found with metric filters or alarms associated.	Monitoring unauthenticated API call read more...	It is recommended that a metric read more... <CIS 1.4.1.3.3 MITRE-ATTACK read more... <CIS 1.4.1.3.3 MITRE-ATTACK read more...
FAIL	medium	cloudwatch	us-east-1	cloudwatch_log_metrics_filter_and_alarm_for_cloudtrail_configuration_changes_enabled	Ensure a log metric filter and alarm exist for CloudTrail configuration changes	38639732453	No CloudWatch log groups found with metric filters or alarms associated.	Monitoring unauthenticated API call read more...	It is recommended that a metric read more... <CIS 1.4.1.3.3 MITRE-ATTACK read more... <CIS 1.4.1.3.3 MITRE-ATTACK read more...
FAIL	medium	cloudwatch	us-east-1	cloudwatch_log_metrics_filter_disable_or_scheduled_deletion_of_customer_created_NATs_Chunks	Ensure a log metric filter and alarm exist for disabling or scheduled deletion of customer created NATs Chunks	38639732453	No CloudWatch log groups found with metric filters or alarms associated.	Monitoring unauthenticated API call read more...	It is recommended that a metric read more... <CIS 1.4.1.3.3 MITRE-ATTACK read more... <CIS 1.4.1.3.3 MITRE-ATTACK read more...
FAIL	medium	cloudwatch	us-east-1	cloudwatch_log_metrics_filter_policy_changes	Ensure a log metric filter and alarm exist for vpc_policy_changes	38639732453	No CloudWatch log groups found with metric filters or alarms associated.	Monitoring unauthenticated API call read more...	It is recommended that a metric read more... <CIS 1.4.1.3.3 MITRE-ATTACK read more... <CIS 1.4.1.3.3 MITRE-ATTACK read more...
FAIL	medium	cloudwatch	us-east-1	cloudwatch_log_metrics_filter_sign_in_without_mfa	Ensure a log metric filter and alarm exist for Management Console sign-in without MFA	38639732453	No CloudWatch log groups found with metric filters or alarms associated.	Monitoring unauthenticated API call read more...	It is recommended that a metric read more... <CIS 1.4.1.3.3 MITRE-ATTACK read more... <CIS 1.4.1.3.3 MITRE-ATTACK read more...
FAIL	medium	cloudwatch	us-east-1	cloudwatch_changes_to_network_route_tables_alarm_configured	Ensure a log metric filter and alarm exist for route table changes	38639732453	No CloudWatch log groups found with metric filters or alarms associated.	Monitoring unauthenticated API call read more...	It is recommended that a metric read more... <CIS 1.4.1.3.3 MITRE-ATTACK read more... <CIS 1.4.1.3.3 MITRE-ATTACK read more...
FAIL	medium	cloudwatch	us-east-1	cloudwatch_log_metrics_filter_for_s3_bucket_policy_changes	Ensure a log metric filter and alarm exist for S3 bucket policy changes	38639732453	No CloudWatch log groups found with metric filters or alarms associated.	Monitoring unauthenticated API call read more...	It is recommended that a metric read more... <CIS 1.4.1.3.3 MITRE-ATTACK read more... <CIS 1.4.1.3.3 MITRE-ATTACK read more...
FAIL	medium	cloudwatch	us-east-1	cloudwatch_log_metrics_filter_security_group_changes	Ensure a log metric filter and alarm exist for security group changes	38639732453	No CloudWatch log groups found with metric filters or alarms associated.	Monitoring unauthenticated API call read more...	It is recommended that a metric read more... <CIS 1.4.1.3.3 MITRE-ATTACK read more... <CIS 1.4.1.3.3 MITRE-ATTACK read more...
FAIL	medium	cloudwatch	us-east-1	cloudwatch_log_metrics_filter_unauthorized_api_calls	Ensure a log metric filter and alarm exist for unauthorized API calls	38639732453	No CloudWatch log groups found with metric filters or alarms associated.	Monitoring unauthenticated API call read more...	It is recommended that a metric read more... <CIS 1.4.1.3.3 MITRE-ATTACK read more... <CIS 1.4.1.3.3 MITRE-ATTACK read more...
FAIL	medium	cloudwatch	us-east-1	cloudwatch_log_metrics_filter_root_usage	Ensure a log metric filter and alarm exist for usage of root account	38639732453	No CloudWatch log groups found with metric filters or alarms associated.	Monitoring unauthenticated API call read more...	It is recommended that a metric read more... <CIS 1.4.1.3.3 MITRE-ATTACK read more... <CIS 1.4.1.3.3 MITRE-ATTACK read more...
FAIL	medium	cloudwatch	us-east-1	cloudwatch_changes_to_vpc_vpn_connections_configured	Ensure a log metric filter and alarm exist for VPN changes	38639732453	No CloudWatch log groups found with metric filters or alarms associated.	Monitoring unauthenticated API call read more...	It is recommended that a metric read more... <CIS 1.4.1.3.3 MITRE-ATTACK read more... <CIS 1.4.1.3.3 MITRE-ATTACK read more...
PASS	medium	iam	us-east-1	iam_support_role_created	Ensure a support role has been created to manage incidents with AWS Support	AWSsupportrole@profreader.us-east-1.amazonaws.com	Support policy attached to role AWSsupportrole@profreader.us-east-1.amazonaws.com	AWS provides a support center read more...	Create an IAM role for support read more... <CIS 1.4.1.3.3 MITRE-ATTACK read more...
FAIL	medium	vpc	us-east-1	vpc_subnet_response_private_public	Ensure all VPC has public and private subnets defined	vpc-093b2310de705b6	VPC vpc-093b2310de705b6 has only public subnets.	Ensure all VPCs have public and read more...	<CIS 1.4.1.3.3 MITRE-ATTACK read more... <CIS 1.4.1.3.3 MITRE-ATTACK read more...
PASS	medium	vpc	us-east-1	vpc_subnet_diffused_ip	Ensure all VPC has subnets in more than one availability zone	vpc-093b2310de705b6	VPC vpc-093b2310de705b6 has subnets in more than one availability zone	Ensure all VPCs have subnets in read more...	<CIS 1.4.1.3.3 MITRE-ATTACK read more... <CIS 1.4.1.3.3 MITRE-ATTACK read more...
FAIL	medium	network-firewall	us-east-1	networkfirewall_rule_all_vpc	Ensure all VPCs have Network Firewall enabled	vpc-093b2310de705b6	VPC vpc-093b2310de705b6 does not have Network Firewall enabled.	Without a network firewall, it read more...	Ensure all VPCs have Network F read more... <CIS 1.4.1.3.3 MITRE-ATTACK read more... <CIS 1.4.1.3.3 MITRE-ATTACK read more...
FAIL	medium	config	us-east-1	config_recorder_all_regions_enabled	Ensure AWS Config is enabled in all regions	38639732453	AWS Config recorder 38639732453 is disabled.	The AWS Config recorder has read more...	It is recommended to enable read more... <CIS 1.4.1.3.3 MITRE-ATTACK read more... <CIS 1.4.1.3.3 MITRE-ATTACK read more...
PASS	medium	cloudtrail	us-east-1	cloudtrail_log_file_validation_enabled	Ensure CloudTrail log file validation is enabled	documentation-ba3b	MultiRegion trail documentation-ba3b log file validation is enabled	Enabling log file validation read more...	Ensure Log file validation is read more... <CIS 1.4.1.3.3 MITRE-ATTACK read more... <CIS 1.4.1.3.3 MITRE-ATTACK read more...

[illegible]

GitHub - prowler-cloud/prowler: Prowler - The Handy Cloud Security Framework

file:///home/kali/cloudvuln/prowler/permissions/output/prowler-output-386397332453-20260312100523.html

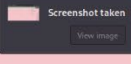
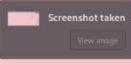
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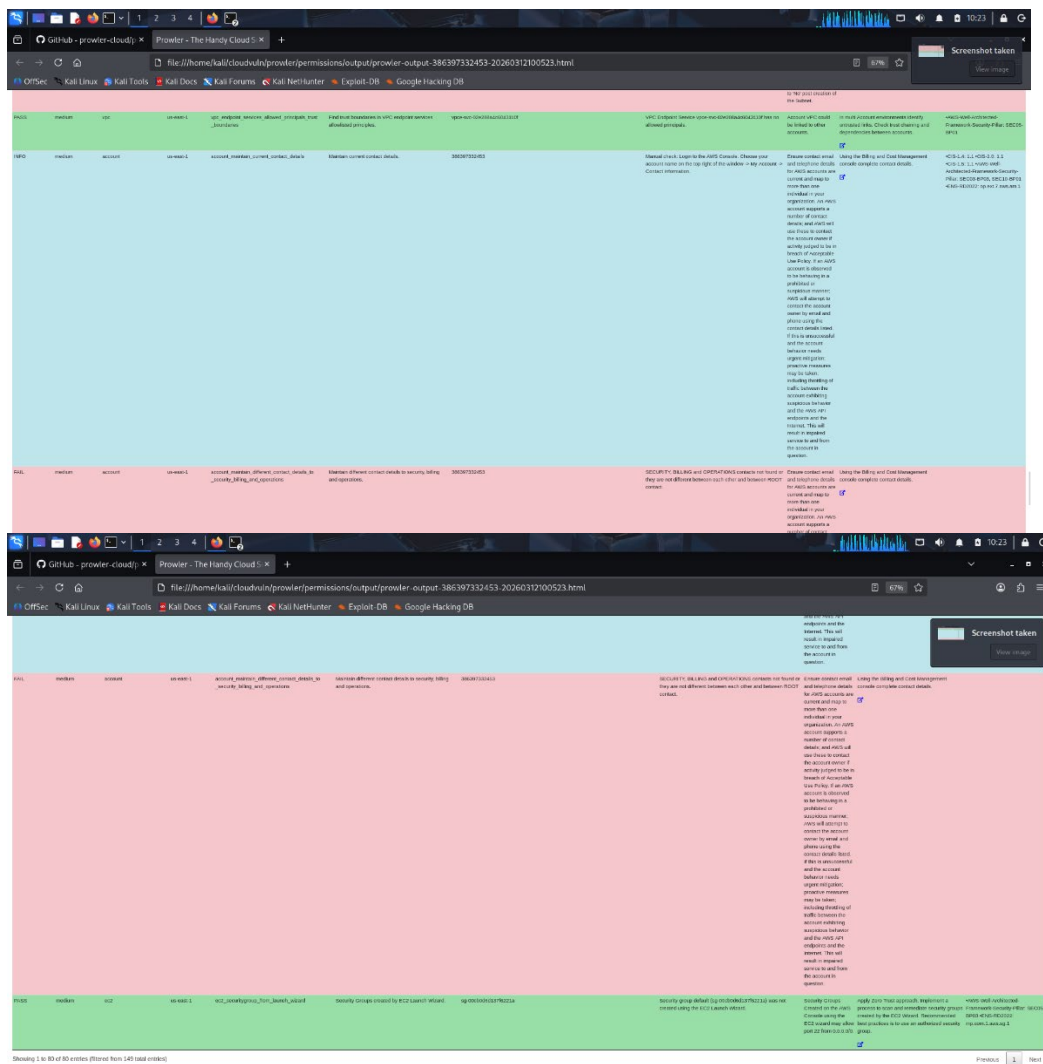
FAIL	medium	iam	us-east-1	iam_user_console_access_unrevoked	Ensure unused user console access are disabled	prowlercloud_user	User prowlercloud_user has logged in to the console in the past 45 days (14 days).	To increase the security of your account, remove IAM user credentials that are no longer needed. For example, when users leave your organization or no longer need AWS access.	First the credentials that they use to access that they are no longer needed. You can remove or deactivate the access keys on the IAM console.
FAIL	medium	iam	us-east-1	iam_user_console_access_unrevoked	Ensure unused user console access are disabled	prowlercloud_user	User prowlercloud_user has logged in to the console in the past 45 days (14 days).	To increase the security of your account, remove IAM user credentials that are no longer needed. For example, when users leave your organization or no longer need AWS access.	First the credentials that they use to access that they are no longer needed. You can remove or deactivate the access keys on the IAM console.
FAIL	medium	iam	us-east-1	iam_user_with_temporary_credentials	Ensure users make use of temporary credentials assuming IAM roles	blu	User blu doesn't have long lived credentials with access to other services than IAM or STS.	As a best practice, use temporary security credentials (not roles) instead of creating long-term credentials like access keys, and don't create AWS account root user access keys.	As a best practice, use temporary security credentials (not roles) instead of creating long-term credentials like access keys, and don't create AWS account root user access keys.
FAIL	medium	iam	us-east-1	iam_user_with_temporary_credentials	Ensure users make use of temporary credentials assuming IAM roles	jean	User jean has long lived credentials with access to other services than IAM or STS.	As a best practice, use temporary security credentials (not roles) instead of creating long-term credentials like access keys, and don't create AWS account root user access keys.	As a best practice, use temporary security credentials (not roles) instead of creating long-term credentials like access keys, and don't create AWS account root user access keys.

file:///home/kali/cloudvuln/prowler/permissions/output/prowler-output-386397332453-20260312100523.html

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FAIL	medium	iam	us-east-1	iam_user_with_temporary_credentials	Ensure users make use of temporary credentials assuming IAM roles	jean	User jean has long lived credentials with access to other services than IAM or STS.	As a best practice, use temporary security credentials (not roles) instead of creating long-term credentials like access keys, and don't create AWS account root user access keys.	As a best practice, use temporary security credentials (not roles) instead of creating long-term credentials like access keys, and don't create AWS account root user access keys.
FAIL	medium	iam	us-east-1	iam_user_with_temporary_credentials	Ensure users make use of temporary credentials assuming IAM roles	prowlercloud_user	User prowlercloud_user has long lived credentials with access to other services than IAM or STS.	As a best practice, use temporary security credentials (not roles) instead of creating long-term credentials like access keys, and don't create AWS account root user access keys.	As a best practice, use temporary security credentials (not roles) instead of creating long-term credentials like access keys, and don't create AWS account root user access keys.
FAIL	medium	vpc	us-east-1	vpc_flow_logs_enabled	Ensure VPC Flow Logging is Enabled in all VPCs	vpc-033b21d0e7056b0	VPC vpc-033b21d0e7056b0: Flow logs are disabled.	VPC Flow Logs provide visibility into network traffic that traverses the VPC and can be used to detect anomalous traffic or misuse during security workflows.	It is recommended that VPC Flow Logs be enabled for packet capture for VPCs.

FAIL	medium	vpc	us-east-1	vpc_subnet_no_public_ip_default	Ensure VPC subnets do not assign public IP by default	subnet-0d78e0970b73064	VPC subnet-0d78e0970b73064 assigns public IP by default	VPC subnet is a part of the VPC having its own rules for traffic. Assigning the Public IP to the subnet automatically (on launch) can accidentally expose the instances within this subnet to internet and should be added to the post-creation of the subnet.	VPC subnets should not allow automatic public IP assignment	
FAIL	medium	vpc	us-east-1	vpc_subnet_no_public_ip_default	Ensure VPC subnets do not assign public IP by default	subnet-0f8a3b3725de35d4	VPC subnet-0f8a3b3725de35d4 assigns public IP by default	VPC subnet is a part of the VPC having its own rules for traffic. Assigning the Public IP to the subnet automatically (on launch) can accidentally expose the instances within this subnet to internet and should be added to the post-creation of the subnet.	VPC subnets should not allow automatic public IP assignment	
FAIL	medium	vpc	us-east-1	vpc_subnet_no_public_ip_default	Ensure VPC subnets do not assign public IP by default	subnet-0978038432be223	VPC subnet-0978038432be223 assigns public IP by default	VPC subnet is a part of the VPC having its own rules for traffic. Assigning the Public IP to the subnet automatically (on launch) can accidentally expose the instances within this subnet to internet and should be added to the post-creation of the subnet.	VPC subnets should not allow automatic public IP assignment	
FAIL	medium	vpc	us-east-1	vpc_subnet_no_public_ip_default	Ensure VPC subnets do not assign public IP by default	subnet-04d6795500b0dc44	VPC subnet-04d6795500b0dc44 assigns public IP by default	VPC subnet is a part of the VPC having its own rules for traffic. Assigning the Public IP to the subnet automatically (on launch) can accidentally expose the instances within this subnet to internet and should be added to the post-creation of the subnet.	VPC subnets should not allow automatic public IP assignment	
FAIL	medium	vpc	us-east-1	vpc_subnet_no_public_ip_default	Ensure VPC subnets do not assign public IP by default	subnet-0a0b96411d55275	VPC subnet-0a0b96411d55275 assigns public IP by default	VPC subnet is a part of the VPC having its own rules for traffic. Assigning the Public IP to the subnet automatically (on launch) can accidentally expose the instances within this subnet to internet and should be added to the post-creation of the subnet.	VPC subnets should not allow automatic public IP assignment	
FAIL	medium	vpc	us-east-1	vpc_subnet_no_public_ip_default	Ensure VPC subnets do not assign public IP by default	subnet-0ac0f0f12804710f	VPC subnet-0ac0f0f12804710f assigns public IP by default	VPC subnet is a part of the VPC having its own rules for traffic. Assigning the Public IP to the subnet automatically (on launch) can accidentally expose the instances within this subnet to internet and should be added to the post-creation of the subnet.	VPC subnets should not allow automatic public IP assignment	
PASS	hardcode	vpc	us-east-1	vpc_endpoint_service_allowed_principal_post_boundaries	Final trust boundaries in VPC endpoint service allowed principles	vpc-en-03a7b64dc05c11cf	VPC Endpoint Service vpc-en-03a7b64dc05c11cf has no allowed principals.	Accounts VPC could be linked to other accounts	In multi-Account environments, identify untrusted links. Check trust chaining and dependencies between accounts	AWS well-architected Framework Security-Pillar, SEC09 Best
INFO	medium	account	us-east-1	account_member_current_contact_details	Maintain current contact details	38630732453	Manual check: Login to the AWS Console. Choose your account name on the top right of the window -> My account -> Contact information	Ensure contact email and telephone details for AWS accounts are current and map to more than one individual in your organization. An AWS account supports a number of contact details, and roles will use these to contact the account owner if actually judged to be in breach of acceptable Use Policy. If an AWS account is determined to be breaching it is prohibited or suspicious manner, AWS will attempt to contact the account owner by email and phone using the	Using the Billing and Cost Management console complete contact details.	<OS 1.4.1.1 <OS 2.0.1.1 <OS 1.5.1.1 <AWS IAM: Authenticated Framework Security-Pillar, SEC09 Best, SEC10 Best, <OS 1.4.1.1 <OS 2.0.1.1



Result	Count
Total Checks	142
Passed Checks	64
Failed Checks	78

Failure rate: **52.35%**

Pass rate: **42.95%**

The summary indicates that a significant number of security checks failed, suggesting that multiple security configurations require improvement.

6.1 Summary Analysis

The Prowler scan summary provides an overview of the distribution of failed security checks across AWS services. The results indicate that the majority of failed checks were associated with Identity and Access Management (IAM), followed by CloudWatch monitoring configurations and VPC network security settings.

This information allows security analysts to prioritize remediation efforts on the services that present the highest number of misconfigurations.

For example:

Service	Failed Checks
IAM	30
CloudWatch	15
VPC	9
EC2	5
CloudTrail	4

The high number of failures within IAM highlights potential weaknesses in identity protection mechanisms such as access control policies and authentication safeguards.

6.2 Detailed Analysis

While the terminal summary provides a high-level overview of security issues, the detailed Prowler report contains individual security findings for each failed control.

These detailed reports includes

- Description of the failed security check
- Severity rating of the issue
- Evidence of the misconfiguration
- Compliance mappings to security frameworks
- Recommended remediation steps

The detailed findings allow security analysts to understand the exact configuration issues within each AWS service and implement appropriate remediation measures.

7. Key Security Findings

7.1 IAM Security Misconfigurations

Several identity security weaknesses were identified within the AWS environment.

Issues detected include:

- IAM users without Multi-Factor Authentication
- Multiple active access keys for a single user
- Weak identity protection mechanisms

These issues increase the risk of credential compromise and unauthorized access to cloud resources.

Affected service: AWS Identity and Access Management

7.2 Monitoring and Logging Weaknesses

The scan revealed missing monitoring configurations in CloudWatch.

Examples include:

- Missing metric filters for authentication failures
- Missing alarms for configuration changes
- Lack of monitoring alerts for suspicious activity

Without proper monitoring, malicious activities within the environment may go undetected.

Affected service: Amazon CloudWatch

7.3 Configuration Monitoring Disabled

The assessment discovered that configuration monitoring was not properly enabled.

Affected service: AWS Config

Without AWS Config, it becomes difficult to track infrastructure changes and detect configuration drift.

7.4 Network Security Exposure

The scan identified potentially risky network access rules.

A network ACL was found allowing inbound access to **RDP port 3389** from the internet.

Affected service: Amazon VPC

This configuration could expose systems to brute-force attacks.

7.5 Encryption Issues

Default encryption for storage volumes was found to be disabled.

Affected services include: Amazon EC2, AWS Glue

This could expose stored data if unauthorized access occurs.

8. Risk Analysis

Most of the identified issues fall into the **medium severity category**, indicating that while they may not immediately compromise the system, they significantly increase the attack surface of the cloud environment.

The concentration of failed checks in identity and monitoring services highlights the need for stronger identity protection and improved security visibility.

9. Remediation Recommendations

To improve the overall security posture of the environment, the following remediation steps are recommended.

Identity Security

- Enforce Multi-Factor Authentication for all IAM users.
- Limit the number of active access keys per user.
- Implement the principle of least privilege for IAM policies.

Monitoring and Logging

- Configure CloudWatch metric filters for security events.
- Enable alarms for suspicious activity.
- Ensure CloudTrail logs are actively monitored.

Network Security

- Restrict inbound RDP access to trusted IP addresses.
- Implement stricter VPC network ACL rules.

Data Protection

- Enable default encryption for EBS volumes.
- Enable encryption for AWS Glue metadata and credentials.

Configuration Monitoring

- Enable AWS Config across all regions.
- Monitor configuration changes for critical resources.

10. Conclusion

The cloud security assessment identified several misconfigurations that could weaken the security posture of the AWS environment. While a portion of the security checks passed successfully, the presence of numerous failed checks indicates that improvements are necessary to align the infrastructure with cloud security best practices.

Addressing the identified issues will significantly strengthen identity protection, improve monitoring capabilities, and reduce potential attack surfaces within the cloud environment.

By implementing the recommended remediation steps, the organization can improve its overall cloud security posture and reduce the risk of unauthorized access or system compromise.